input



File: power_input.kicad_sch

buckboost



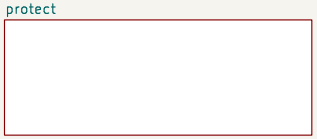
File: buckboost.kicad_sch

mcu



File: mcu.kicad_sch

protect



File: protect_sense.kicad_sch

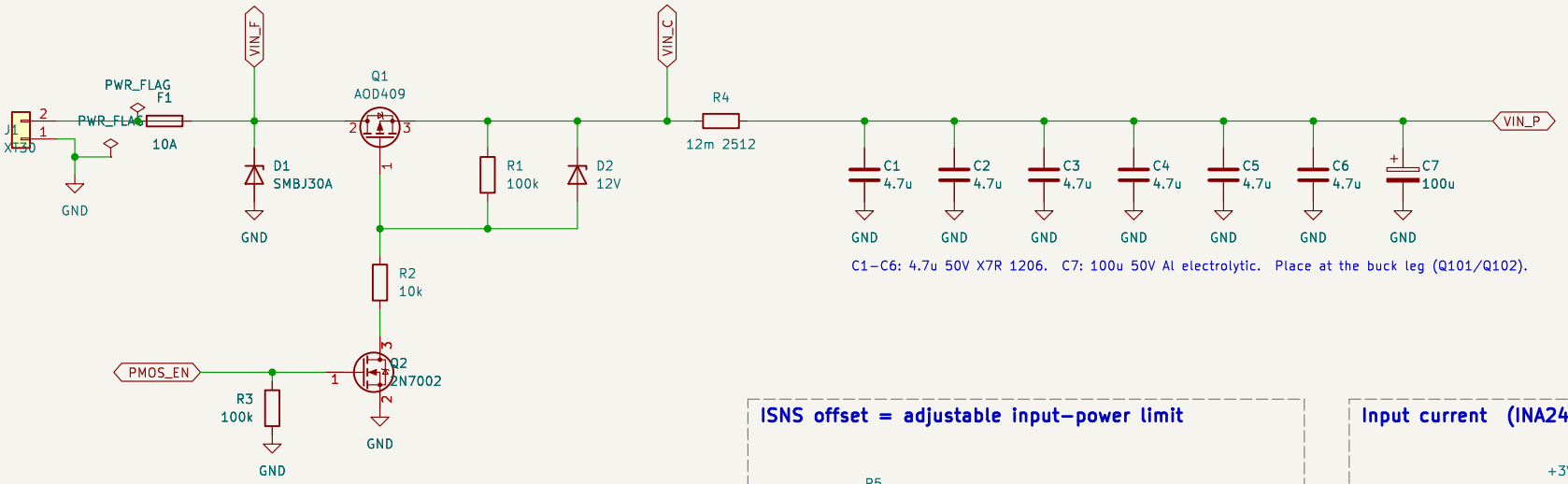
Sheet: /
File: control_board.kicad_sch

Title: Rev B control board (root)

Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/5

1/4 Power input (PM chassis output 22–26 V)

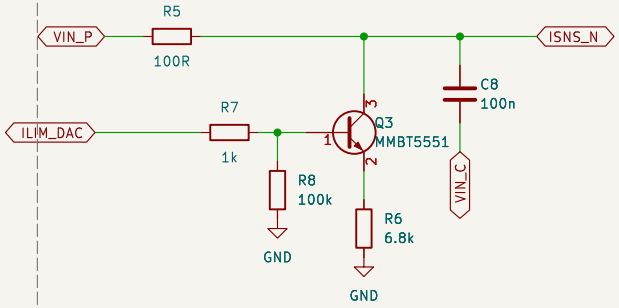
Q1 blocks back-feed from converter/bank into the PM output (body diode conducts input -> board only). Q1 stays OFF unless PMOS_EN is high.
R4 = LM5176 input shunt (ISNS limit 50 mV -> 4.2 A). Kelvin-route ISNS+ (VIN_C), ISNS- (via R5) and the INA240 inputs from the R4 pads.



C1–C6: 4.7u 50V X7R 1206. C7: 100u 50V Al electrolytic. Place at the buck leg (Q101/Q102).

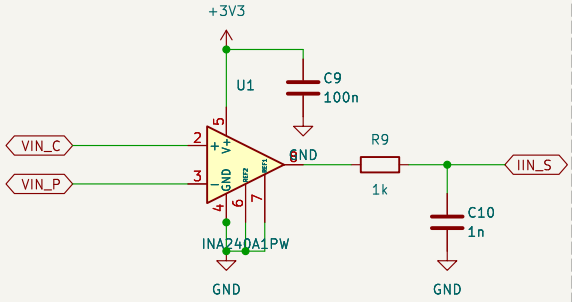
PMOS_EN (sheet 4) = MCU EN_PMOS AND no input OV. R3 keeps Q1 off at power-up.

ISNS offset = adjustable input-power limit



I_ofs = (V_DAC − 0.65 V)/R6, max ~0.4 mA -> +40 mV on ISNS
=> limit 4.2 A ... ~0.8 A. MCU closes the loop with IIN_S.

Input current (INA240A1, G=20 -> 0.24 V/A)



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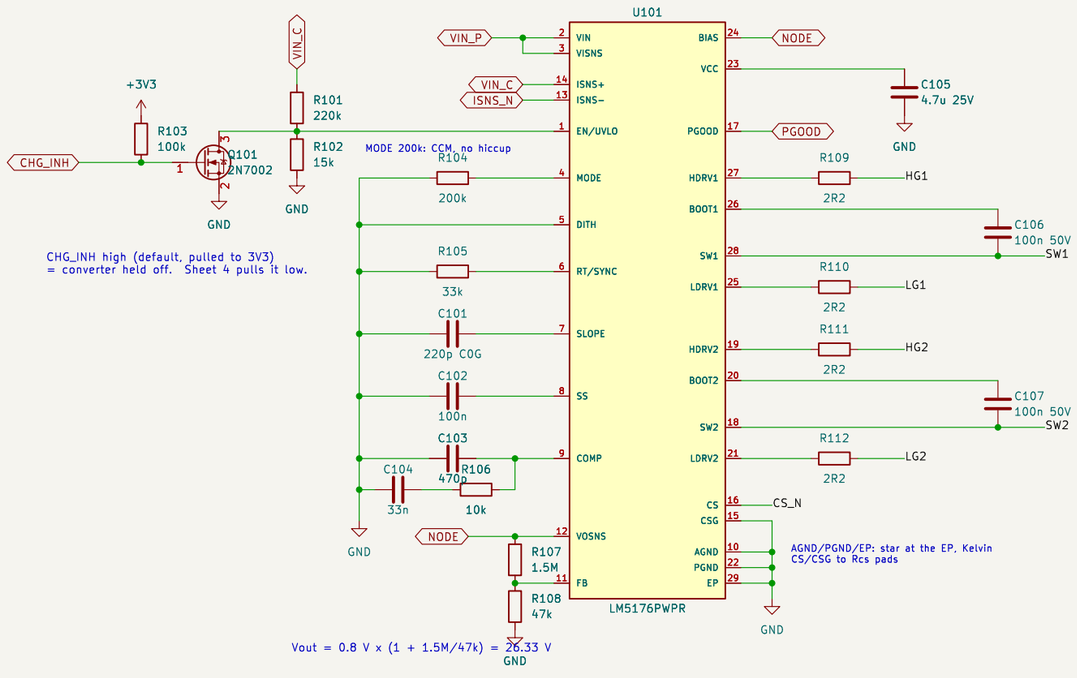
Sheet: /input/
File: power_input.kicad_sch

Title: Rev B control board – 1/4 power input

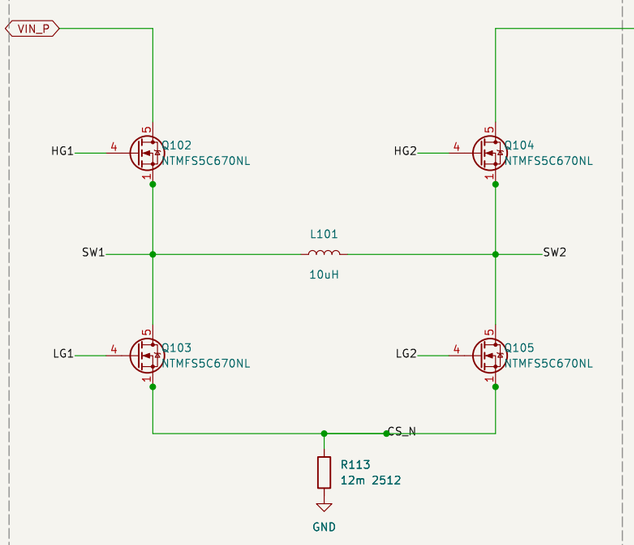
Size: A4	Date:	Rev: B–v3
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 2/5

2/4 LM5176 4-switch buck-boost charger (VIN_P 22-26 V -> NODE 0-26.2 V, ~100 W)

NODE = chassis output and capacitor bank (via CM01). fsw ~248 kHz (RT 33k), Vout 26.3 V (1.5M/47k), UVLO ~19.3 V (220k/15k), CCM, hiccup disabled (MODE 200k).
Power FETs: 60 V class (input TVS SMBJ30A clamps up to ~48 V). Gate nets HG1/LG1/HG2/LG2 and SW1/SW2 are local labels; keep gate loops short in layout.

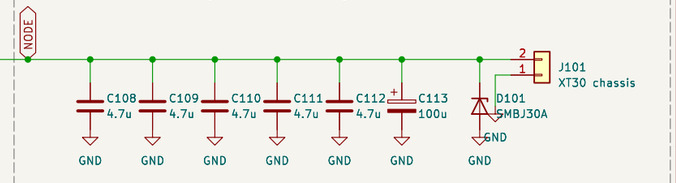


Power stage

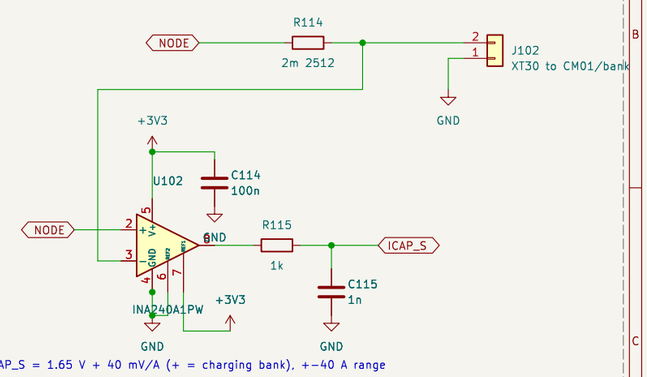


Current limits over Rcs 12 mR (min/typ/max):
 buck valley 66/80/94 mV = 5.5/6.7/7.9 A, boost peak 100/120/140 mV = 8.3/10/11.8 A
Worst L peak ~12.5 A (valley max + ripple, L -20% tol, -14% at 12 A) < Isat 17 A
CSLOPE = L x 2 uS / (Rcs x 5) = 333 pF at 10 uH. 220 pF C0G gives
 1.0-1.9x dead-beat slope over L = 6.9-12 uH (datasheet: stable above 0.5x)

Output node NODE (chassis + bank)



C: 4.7u 50V X7R 1206 at the boost leg.
D: clamps regen spikes if the bank is cut off.



ICAP_S = 1.65 V + 40 mV/A (+ = charging bank), +-40 A range

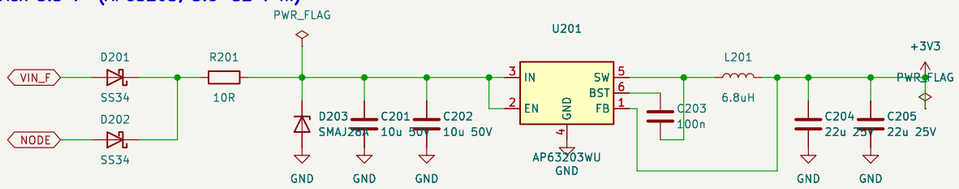
NAGOYA SHACHIHOKO

Sheet: /buckboost/		
File: buckboost.kicad_sch		
Title: Rev B control board - 2/4 buck-boost charger		
Size: A3	Date:	**Rev: B-v3**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/5

3/4 Aux 3.3 V supply, MCU (STM32G431CBT6), CAN, referee UART

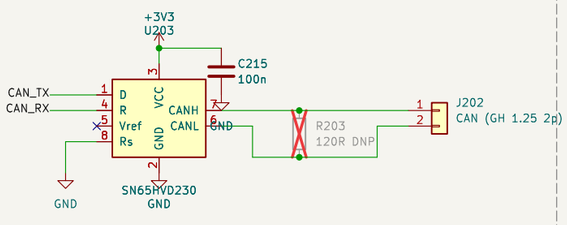
3V3 is fed from PM input OR the bank (NODE), so the MCU stays alive while the robot is unpowered (bank drain ~0.15 W). MCU runs only the slow outer loop.

Aux 3.3 V (AP63203, 3.8–32 V in)

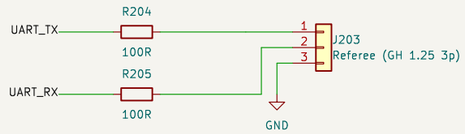


D: SS34 (40V 3A). R 10R limits SMAJ28A current so AUX_IN stays below the AP63203 35 V abs max.

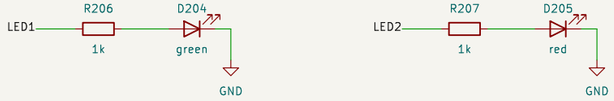
Robot CAN (SN65HVD230, 1 Mbps)



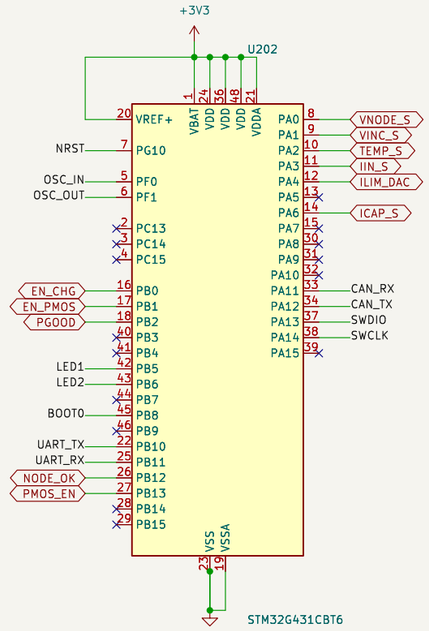
Referee UART (3.3 V TTL)



Status LEDs

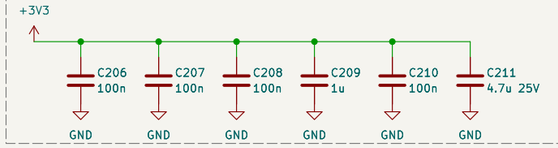


MCU

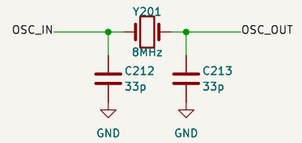


PA4 = DAC1_OUT1 (ILIM_DAC), PA6 = ADC2_IN3, PA11/12 = FDCAN1, PB10/11 = USART3.

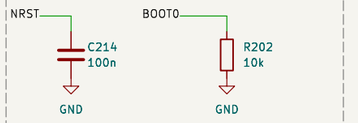
MCU decoupling (one 100n per VDD pin, 1u at VDDA)



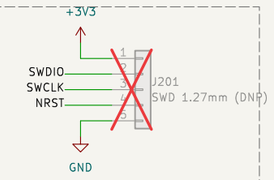
8 MHz crystal (FDCAN clock)



Reset / BOOT0



SWD



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Sheet: /mcu/

File: mcu.kicad_sch

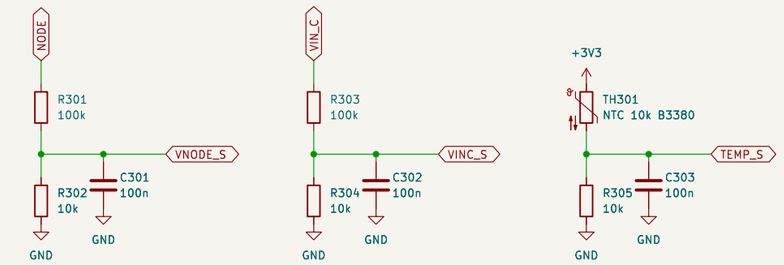
Title: Rev B control board — 3/4 MCU and aux power

Size: A3	Date:	Rev: B–v3
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/5

4/4 Sensing, hardware over-voltage protection, converter enable logic

The converter runs only if PMOS_EN (MCU EN_PMOS and no input OV) AND NODE_OK (no node OV) AND EN_CHG (MCU). Everything defaults to OFF. The comparators work without the MCU.

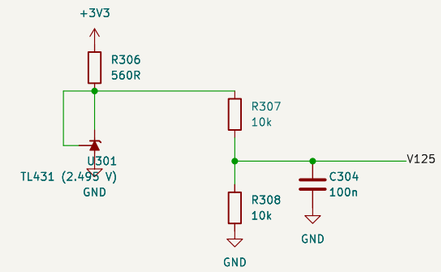
Voltage / temperature sensing (ADC)



27 V -> 2.45 V

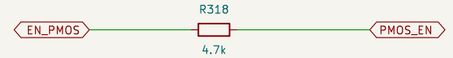
26 V -> 2.36 V

1.25 V comparator reference

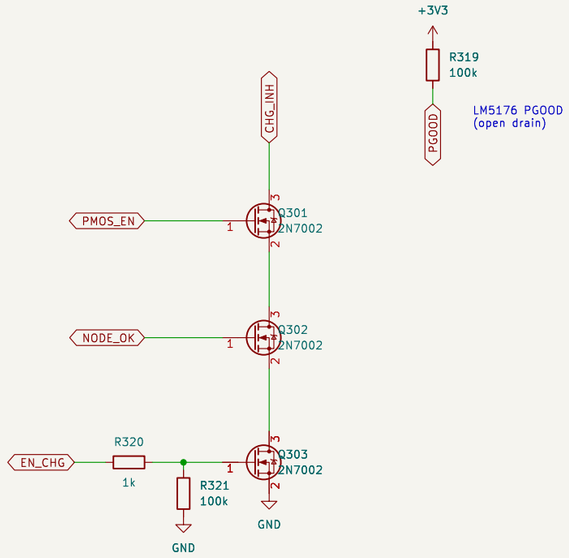


1.2475 V: below the LM393 input range limit (Vcc − 1.5 V).

Converter enable logic (default OFF)

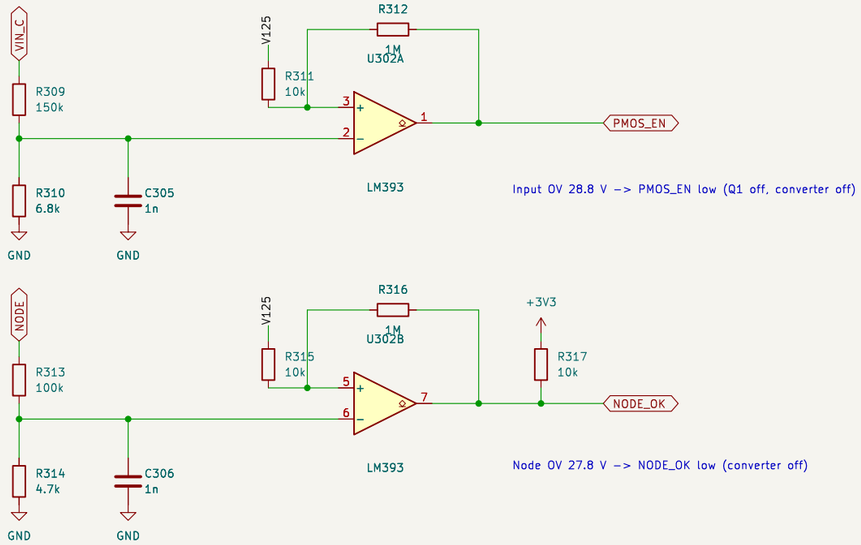


MCU sources PMOS_EN through 4.7k; comparator A sinks it on input OV.
100k pull-down at Q2 (sheet 1) keeps it low at reset.



Qa.Qb.Qc all ON -> CHG_INH pulled low -> LM5176 EN released.
Any of them OFF -> CHG_INH high (100k to 3V3, sheet 2) -> converter off.

Hardware over-voltage comparators (LM393, open collector)



Input OV 28.8 V -> PMOS_EN low (Q1 off, converter off)

Node OV 27.8 V -> NODE_OK low (converter off)

Hysteresis: 10k/1M on the + input, about 0.7 V at the monitored rail. 1n filters on the dividers.

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Sheet: /protect/		
File: protect_sense.kicad_sch		
Title: Rev B control board — 4/4 sensing and protection		
Size: A3	Date:	**Rev: B-v3**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/5